Music 451C

The repo for the 2026 apparatus:

https://cm-gitlab.stanford.edu/matt/dual-pad_eeg

The old (for Alex Han) dual-pad_improv patch:

https://cm-gitlab.stanford.edu/matt/dual-pad_improv

Blocks (set 1 and 2 - March 2)

2026/03/02
Strategy of Joint Improvisation

Complementary (Two-part):
Consider the analogy to Yin and Yang. In this condition, each player will bring a distinct character, not 6 randomly different phrases.

Continuity (One-part):
Starter begins with some idea, and Joiner will continue developing that idea (not repetition), and with turn-taking both players will continue this development.

2026/03/05

Timbre and Deviant Assignment are updated

We will assign an instrument category to a person (Timbale or Conga). And, the deviant-type #1 (in the blockfile, it's 1) is self category, deviant-type#2 (in the blockfile, 3) is coming from the partner's category.

Pad1:
Timbale: 1,2,3,4 (assigned to pad),5,6,7 (higher pitch for deviant, corresponding to the number 1,2,3 for the variation, no 'mod 4' needed)

Pad2:
Conga:1,2,3,4 (assigned to pad), 5,6,7 (higher pitch for deviant, corresponding to the number 1,2,3 for the variation, no 'mod 4' needed)

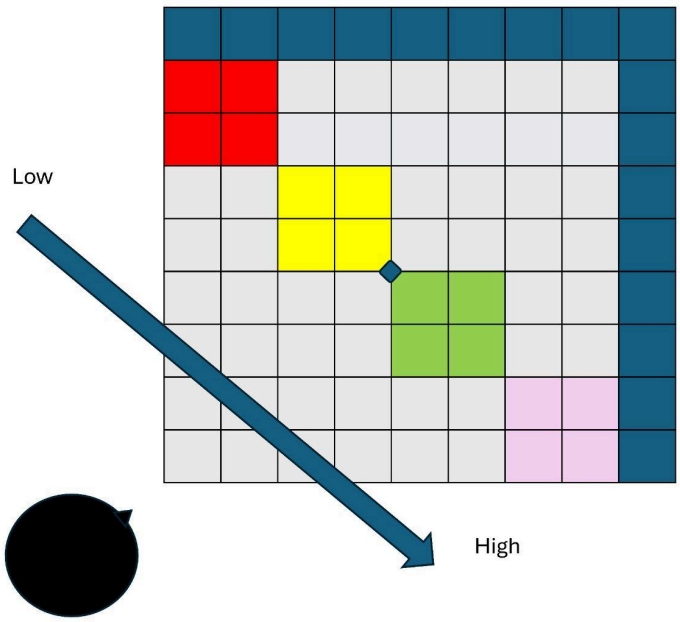
Blockfiles stay same.

Soundfile naming convention would be:

Pad Design renewed

Timba1, Timba2, Timba3, Timba4, Timba5, Timba6, Timba7 (from lower pitch to higher pitch)

Conga1, Conga2, Conga3, Conga4, Conga5, Conga6, Conga7 (from lower pitch to higher pitch)



2026/02/03 Takako's note

Task:

- Alternate Phases: 1,2,3,4,5,6,+ Unison
- 100 BPM (4/4) Metronome count-in of 4
- One phrase should include at least five notes
- Deviant note (fourth or fifth note) would happen once per person per trial (Starter (1,3,5) Joiner (2,4,6)) -> Block files
- Successive phrases can overlap for a certain time window (say, the phrase 1 goes on longer than four-beat worth duration (2.4 sec at 100BPM) and spill over, say, 1.0 beats (let's call this as the overlap window = 0.6s) into the phrase 2 if Player 2 already started, but the next person could come in really after the player 1 is done done, and that silence could be also say a certain duration, say 0.6 s again (let's call this as the gap window = 0.6s), then the timer of the phrase 2 can start when from the phrase 1 onset, 2.4 + 0.6 = 3.0 = phrase 2 onset.
- Unison (Starter: quarter-quarter-(then Joiner join) eighth-eighth-quarter)
    - We can score this with a finite state machine, so that the timing doesn't count as long as the sequence of notes is correct with two from starter, then three "unison" (for which either can be first, but then the other has to

be next, like the unison notes in the Chafe clapping pattern from Aury's experiment)
    - Also, having recognized this successful ending phrase, we retroactively know the time of its downbeat (namely the first of starter's two solo notes) for purposes of judging the duration of the 6th phrase.
    - Unsuccessfully ended trials will have to be aborted manually by the experimenter, or we can say that after joiner plays in phrase 7, any pause (in which neither player plays) above a certain duration triggers the trial's end (in failure).
- If the overlap window or gap window became more than the designated value, Error trial (80)
- If any phrase is too short (<1.8 sec (2.4-0.6) or too long (>3.0 sec (2.4+0.6), Error trial
- If a phrase has four notes or less, then Error trial
- In other words, if there is a time segment that contains only one player's notes (5 or more) that goes for 1.5-3.3 sec, then it qualifies as a phrase
- The next player can start as late as 1.5 sec after the previous player's starting note and could go as long as 3.3 sec
- Previous player can play the last note then the next player could start 0.9 sec after that last note.
- (there might be some logic missing….)

(2026-02-11) New idea on scoring correctness of turn-taking:

- Each trial has six "turns": (metronome), 1:starter, 2:joiner, 3:starter, 4:joiner, 5:starter, 6:joiner, (ending = 7:starter + 8:joiner)
- Each turn's nominal length is four beats (same as metronome)
- Each turn begins with the first note played; this is not necessarily on the musical downbeat
- The trial (= "piece") ends if both players don't play any notes for at least 2.5 beats
- The turn ends after the player doesn't play another note for at least 2.5 beats
    - Therefore each player's division of notes into turns is *independent* - it doesn't matter what I play or not during your 2.5 beat gap.
- Each turn must contain at least 5 notes
- The next turn must begin no sooner than 3 beats (3 x 0.6=1.8s) after the beginning of this turn
- The next turn must begin no later than 5 beats (5 x 0.6 = 3.0s) after the beginning of this turn

- The turn must end with a successful performance (per finite state machine) of the final rhythmic pattern
- After the end of the final rhythmic pattern, there must be at least 300ms (or 500?) with no notes from either subject

(2026-02-18) Number crunching

- 1-bar metronome (4 count-ins) + 6 phrases of turn-taking + 1-bar Unison phrase = total of 8 bars per trial
- Inter-trial-interval (from the end of the previous trial to the first metronome count-in) 2.5-3.5 sec
- If 100BPM, 1 beat = 0.6 sec, 32 beats=19.2
- 2 practice trials + 27 test trials = 29 trials per block
- 1 block takes 9.5 min (29 x (19.2+3.0=22.2 ITI) sec = 643.8sec = 10.73 minutes)
- (If 110 BPM, 1 beat = 0.54.5 sec., and the entire block will be shorter by 100/110, so 10.73 * 100/110 = 9.754 min)

  Fusion
  - P1start and fusion, then 2 blocks (54 test trials) contain 27 dev1, 27 dev
  - P2 start and fusion, then 2 blocks(54 test trials) contain 27 dev1, 27 dev2

  Complementary
- P1 start and complementary, then 2blocks 27 dev1, 27 dev2
- P2 start and complementary, then 2 blocks 27 dev1, 27 dev2

FRN each of the deviant type (dev1 and dev2) has 54 trials per condition (fusion or complementary)

Alpha
P1start and fusion, 108 dev-free phrases per player
P2start and fusion, 108 dev-free phrases per player
P1start and complementary, 108 dev-free phrases per player
P2start and complementary, 108 dev-free phrases per player

Recording goes through 8 blocks in total

- 10min x 8 blocks = 80 min
- Capping two people = 60 min
- Practice and instruction = 15 min

- Questionnaire = 15min
- Total 80+60+15+15=170min (2 hours 50 min)

If we employ 120 BPM 1 beat becomes 0.5 sec (instead of 0.6)
170min * ⅚ = 141 minutes (2 hours 21 min)

If we employ 110 BPM 1 beat becomes 0.545 sec
170min * 0.545/0.6=154 minutes (2 hours 34 min)

Trigger codes

- First metronome tock's trigger code encodes the condition: **200, 201, 202, 203, 204**
    - Practice/Regular
    - P1start vs. P2start
    - The strategy
- Subsequent metronome tocks: (it can be **233** again, x 3 times)
- Starter, Phrase 1: start the note with the trigger as **11**, and continue 11 for each note as long as this player continues to play
- If deviant note is inserted in Phase 1: **14** or **15** if "self timbre", or **16** or **17** if "other timbre"
- Joiner, Phrase 2: **21**, 21, (Dev: **24** or **25** or **26** or **27**)
- Starter, Phrase 3: **31**, 31,...(Dev: **34**, **35, 36, 37**)
- Joiner, Phrase 4: **41**, 41,...(Dev: **44**, **45, 46, 47**)
- Starter, Phrase 5: **51**, 51,...(Dev: **54**, **55, 56, 57**)
- Joiner, Phrase 6: **61**, 61,...(Dev: **64**, **65, 66, 67**)
- Starter, Phrase 7, first two beats: **71**,**72** (two notes alone)
- Starter, Phrase 7 the remaining three notes: **73**, **74**, **75**
- Joiner, Phrase 7 the remaining three notes: **76**, **77**, **78**
- Trial outcome - when the trial ends, 1.5s after final played note (legacy codes):
    - **79**: aborted
    - **80**: error
    - **81**: success
- If two notes are played within 5ms, Curry won't be able to receive their separate trigger codes in realtime, because of its limit in needing to "sample" the trigger codes with the same clock as the EEG electrodes. So the patch will send the *first* trigger code normally, then instead of sending the second trigger code it will

wait a short amount of time and then send the special code meaning that this happened (and you'll have to look in the logfile to see the second trigger): **150**

Blockfile columns:

1. Practice (0) vs. regular trial (1)
2. * which collaboration style (Two-parts (2) vs. One-part(1))
3. * Which player (from the perspective of MIDI keyboards 1 and 2) starts? (P1start or P2start)
4. Starter's deviant: which phrase (1 or 3 or 5)
5. Starter's deviant: which note (4 or 5)
6. Starter's deviant: timbre choice (plus 1, 2, or 3 (mod 4))
7. Starter's Each (starter's own set=1 or "**the other party"** set = 3)
8. Joiner's deviant: which phrase (2 or 4 or 6)
9. Joiner's deviant: which note (4 or 5)
10. Joiner's deviant timbre choice (plus 1, 2, or 3 (mod 4))
11. Joiner's deviant timbre type (joiner's own set=1 or "**the other party"** set = 3)
12. Trigger code of first metronome (encoding trial condition: 200 = practice trial,
    201=P1start*Two-part,
    202=P2start*Two-part,
    203=P1start*One-part,
    204=P2start*One-part)

Columns marked * should be consistent throughout a block

What we need: 54 lines of 201, 54 lines of 202, 54 lines of 203, 54 lines of 204

1. Spreadsheet view:

https://docs.google.com/spreadsheets/d/18ql1si-E0qo-69iTsnYKLE5v1blge163bXGuJqfzabI/edit?gid=0#gid=0

Deviant notes

- the played note is one of the four. So, the other timbre would be one of the remaining three. But, which one cannot be prescribed beforehand, because all phrases all notes are improvised. So Max has to randomly select. To balance how many timbre deviant goes to so-and-so timbre, maybe you can say, for this deviant, pick the clock-wise next one, or for the next, pick the counter-clock-wise

next one, etc. You just cannot control what timbre note is played in a given deviant position.

- 2/13/26
    - Takako and Udbhav had meeting
    - We will have two types of deviants
    - If partner 1 intended to play a tom
    - Deviant 1 would be any of the other 3 instruments available on partner 1's pad
    - Deviant 2 would be any of the other 3 instruments but from the third party drum set
    - We will have 3 drum sets with the same types of instruments - partner 1, partner 2, and third party set
    - Each drum set have 4 instruments
    - Corresponding instruments on each set will have similar timbres to each other
        - Eg - the three toms will have similar timbres to each other
    - We will identify timbre similarity with a preliminary rating task amongst ourselves
        - Matteo and Calvin will bring 6 possible sets consisting of 4 distinct percussive instruments
        - Each of the 4 instruments per set should occupy different frequency bands
        - However, there shouldn't be a consistent frequency relationship between corresponding instruments across the 2 drums sets
            - Eg - if the tom is higher frequency in set 1, the hi hat should be lower frequency in set 2
            - Should not have a case where all four instruments are higher frequency in set 1 than in set 2
        - We will rate the similarity of all the combinations of pairs of the same instrument on a scale of 1 to 10
        - 6 sounds per instrument category - 15 pairs within each category - 4 seconds per pair - 4 minutes per round, each rater does 2 rounds - 8 minutes total
        - Within each instrument category, we find the three sounds with the highest similarity
        - These instruments used as the 3 drums sets
    - Why are we doing this 2 types of deviants

        - D1 is interesting because you are hearing a sound that you could have produced but is not the sound you were expecting.
        - This may throw off self FRN in the fusion condition more than the complimentary condition because the threshold of forgiveness of mismatch sound would be less in the fusion condition
        - This deviant is still in the realm of actions that you as a improviser could take
        - D2 is interesting because both partners are hearing a sound that is not within either of their realm of possibility
        - The interference from the third party would perceived similarly between both participants because the deviant is outside the actions they can take
        - Self and other shouldn't be as different as in the case with deviant 1

Procedure to start the apparatus (2026/02/25)

1. Cable connections
    a. powered USB hub (D-link) for two drum pads
    b. Macbook needs two USB adapters
2. Equipment to power on
    a. Two loud speaker stands in the booth
    b. Yamaha mixer (MOTU output 7/8 on, with Stereo (red) button on)
    c. MOTU on
    d. Desk speaker on
    e. Curry computer (EEGACQ) on (lower side Dell PC) (account: neuromusic lab, music2brain)
    f. SynAmps RT (Neuroscan) two units should be on (see blue lights in front)
    g. Launch Curry 8 (see the USB dongle in the front lit up a red light)
    h. Headbox #1 and #2 for eeg should indicate numbers
3. Boot the MacBook Air (account: neuromusic lab, music2brain) and confirm launchpads' many usb connections
4. Open the folder "dual-pad_eeg"
    a. Open the patch +Drum-Duo-EEG.maxpat inside the "max" folder
    b. Confirm which identical launchpad is assigned to 1 and 2
    c. Drag the folder of blockfiles (e.g., blockfiles/firstbatch-feb25) onto the patch (on the red rectangle "drop folder of blockfiles here")
    d. This automatically creates a log file in the same folder as the block files, prefix "2PEEG-log-" (for "dual pads EEG"), then the exact time of reading the blockfiles as YYMMDDHHMMSS, with ".coll.txt" file extension(s).

e. Choose which block to attempt; click large button “actually start next block”
f. Confirm stim codes
    i. See the yellow data light on the beige box
    ii. Stim switcher should be on “B Arduino”

====== Below is Archival =======

Checklist for Decision making

For Max/MSP patch

- Blockfiles (definition of trials and conditions)
    - Each line fully specifies one trial - we design the exact fields/columns
    -
- Trigger codes
- Task (music)
    - Metronome (BPM)
        - 100 BPM was somewhat better
    - Time signature
        - 4/4
    - P1start / P2start
    - Do the percussion sounds change? (Different samples, different pitches, different roles…)
        - We want slightly different timbre between Player1 and 2 but not specific pitch intervals
    - Definition of Improvisation (# of notes per phrase, can be within min/max)
        - See the notes above
    - Codes or Bass accompaniment (whole or half notes)
        - Decided not to have any backgrounds

    - Should subjects align at turn-taking boundaries by playing that downbeat in (quasi-)unison?
        - No, we allow overlaps and underlaps
- Deviants
    - We want the sound to be something that could have been played, but wasn't
    - Which dimensions?
        - Timbre (which of the four pads) seems the most straightforward.
        - Time would have to be delay, and there could be ambiguity with the timing relationship between the next-played regular note versus the sounding of the delayed deviant note
        - Velocity (if allowed) is a little complicated too. The simplest "opposite" would be 127-x, but this has very little effect at mezzo values versus increasing effect towards pppp and ffff. So to avoid that, it would be something like thirds of the range, where pp becomes ff, ff becomes mf, and mf becomes pp.
- Conditions
    - Improv-imitative, Improv-complementary
        - I am thinking
        - 'Imitative' could mean 'one part or line played by two people'
        - 'Complementary' could mean 'two parts like dialogue (call-and-response) but bringing two things alternating'

Decisions from Feb 18 2026

OK not to have a separate "practice mode"

BPM 100

Matt's Questions for Feb 23 2026

Are the drum sounds properly assigned to the three sets?

Propose "trial ending gap" of 2.5 (or ?) beats duration - required after completing the ending figure for trial to be a success. (Otherwise a shorter required pause after the end of the FSM detection?)

- Should it be more permissive right after the metronome (maybe count from the implied fifth tock "downbeat" instead of from the fourth tock?)

(email with Takako): after trial ends, ignore notes (same as "practice") during the next trial's metronome count-in; subjects' played notes "don't count" until the moment of the final (fourth) metronome tock.

"Timing was so close that we dropped trigger ___ and later sent ___"

- We "expect" this in the final three notes of the ending figure
- Also subjects might play improv notes as "chords" (2+ pads within 1 ms)

Do we now need a "pause block" feature?

Existing Apparatus Features Easily Available

(the "Chipotle" options)

- Experimenter-facing dashboard showing "everything" the subjects and patch are doing
    - Abort Current Trial button (then pause and start next)

    - Manually select any block
        - preview conditions of block's first trial
        - Then manually trigger start of block
- Drag-and-drop any folder of blockfiles from the Finder onto the patch
- Practice mode for subjects to learn the instruments
- Trigger codes output any time via USB "serial" / Arduino;
    - special handling for codes too close together (unison notes)
- Logging: time-stamp and whatever fields for each kind of event (played note, trigger code, trial outcome…)
- Metronome generation: any #, BPM, "tock" sound file
- A score showing what one or both players does for certain trials
    - Represented as a text file of integers
    - Software can follow, so as to make certain notes be deviants
        - Or it can just count sequential notes and make the nth deviant
    - There can be an ending unison section after a mostly-alternating trial
- Rating of success vs. fail trials (according to certain deterministic criteria)
    - maintaining each block's "to do list" so that subjects have to "make up" failed trials to end with the correct number of *successful* trials.
    - Special handling of practice trials, where failures must be made up immediately
    - "Infinite forgiveness" test mode
- Synthesized voice saying "Block [one] complete" after each block
    - built-in break, waiting for experimenter to start next block manually, e.g., after confirming the next block's task conditions
- Automatic random-within-range pause between trials and starting the (metronome for the) next.
- Open a web site (e.g., survey)
- Custom simple visualization for subjects (requires patching video to booth)
- Automatic bassline accompaniment, either triggered synchronously when the subject plays a certain note, or based on a (perfect) internal metronome.
- Synthesis can be anything you can do in Max/MSP, typically straightforward sample playback triggering